

# Minimal CPS Translation

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In this note, we review one of the standard translations to continuation-passing style.

**Definition 1** (Syntax).

$$\begin{aligned} t &\stackrel{\text{def}}{=} x \mid \lambda x. t \mid t t \\ v &\stackrel{\text{def}}{=} \lambda x. t \end{aligned}$$

**Definition 2** (Call-by-name — weak, closed, small-step semantics).

$$\frac{}{(\lambda x. t) s \rightarrow t\{x := s\}} \beta \quad \frac{t \rightarrow t'}{t s \rightarrow t' s} \mu$$

**Definition 3** (Call-by-value — weak, closed, big-step semantics).

$$\frac{}{\lambda x. t \Downarrow \lambda x. t} \text{e-lam} \quad \frac{t \Downarrow \lambda x. t' \quad s \Downarrow v \quad t'\{x := v\} \Downarrow v'}{t s \Downarrow v'} \text{e-app}$$

**Definition 4** (CPS translation).

$$\begin{aligned} x^\circ &\stackrel{\text{def}}{=} \lambda k. k x \\ (\lambda x. t)^\circ &\stackrel{\text{def}}{=} \lambda k. k (\lambda x. t)^\bullet \\ (t s)^\circ &\stackrel{\text{def}}{=} \lambda k. t^\circ (\lambda x. s^\circ (\lambda y. x y k)) \\ (\lambda x. t)^\bullet &\stackrel{\text{def}}{=} \lambda x. t^\circ \end{aligned}$$

**Lemma 5.**  $t\{x := v\}^\circ = t^\circ\{x := v^\bullet\}$

*Proof.* By induction on  $t$ . □

**Theorem 6.** *If  $t \Downarrow v$  then  $t^\circ K \rightarrow K v^\bullet$  for any term  $K$ .*

*Proof.* By induction on the derivation of  $t \Downarrow v$ .

- e-lam:

$$(\lambda x. t)^\circ k = (\lambda k. k (\lambda x. t)^\bullet) K \rightarrow K (\lambda x. t)^\bullet$$

- e-app:

$$\begin{aligned} (t s)^\circ K &\stackrel{\text{def}}{=} (\lambda k. t^\circ (\lambda x. s^\circ (\lambda y. x y k))) K \\ &\rightarrow t^\circ (\lambda x. s^\circ (\lambda y. x y K)) \\ &\rightarrow (\lambda x. s^\circ (\lambda y. x y K)) (\lambda x. t')^\bullet && \text{by i.h. on the first premise} \\ &\rightarrow s^\circ (\lambda y. (\lambda x. t')^\bullet y K) \\ &\rightarrow (\lambda y. (\lambda x. t')^\bullet y K) v^\bullet && \text{by i.h. on the second premise} \\ &\rightarrow (\lambda x. t')^\bullet v^\bullet K \\ &\rightarrow t'^\circ \{x := v^\bullet\} K \\ &= t'^\circ \{x := v\}^\circ K && \text{by Lem. 5} \\ &\rightarrow K v'^\bullet && \text{by i.h. on the third premise} \end{aligned}$$

□

# 1 Types

**Definition 7** (Simply typed  $\lambda$ -calculus).

$$\begin{aligned} A &::= \perp \mid \alpha \mid A \rightarrow A & \neg A &\stackrel{\text{def}}{=} A \rightarrow \perp \\ \Gamma &::= \emptyset \mid \Gamma, x : A \end{aligned}$$

Rules:

$$\frac{}{\Gamma, x : A \vdash x : A} \text{t-var} \quad \frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash \lambda x. t : A \rightarrow B} \text{t-lam} \quad \frac{\Gamma \vdash t : A \rightarrow B \quad \Gamma \vdash s : A}{\Gamma \vdash t s : B} \text{t-app}$$

**Definition 8** (CPS translation on types).

$$\begin{aligned} \perp^\circ &\stackrel{\text{def}}{=} \perp & \alpha^\circ &\stackrel{\text{def}}{=} \alpha & (A \rightarrow B)^\circ &\stackrel{\text{def}}{=} A^\circ \rightarrow \neg\neg B^\circ \\ \emptyset^\circ &\stackrel{\text{def}}{=} \emptyset & (\Gamma, x : A)^\circ &\stackrel{\text{def}}{=} \Gamma^\circ, x : A^\circ \end{aligned}$$

**Theorem 9.** If  $\Gamma \vdash t : A$  then  $\Gamma^\circ \vdash t^\circ : \neg\neg A^\circ$ .

*Proof.* By induction on the derivation of  $\Gamma \vdash t : A$ .

- t-var:

$$\frac{\frac{\Gamma^\circ, x : A^\circ, k : \neg A^\circ \vdash k : \neg A^\circ}{\Gamma^\circ, x : A^\circ, k : \neg A^\circ \vdash k x : \perp}}{\Gamma^\circ, x : A^\circ \vdash x^\circ = \lambda k. k x : \neg\neg A^\circ}$$

- t-lam:

$$\frac{\frac{\Gamma^\circ, k : \neg(A^\circ \rightarrow \neg\neg B^\circ) \vdash k : \neg(A^\circ \rightarrow \neg\neg B^\circ)}{\Gamma^\circ, k : \neg(A^\circ \rightarrow \neg\neg B^\circ) \vdash \lambda x. t^\circ : A^\circ \rightarrow \neg\neg B^\circ} \text{i.h. + weakening} \quad \frac{\Gamma^\circ, k : \neg(A^\circ \rightarrow \neg\neg B^\circ), x : A^\circ \vdash t^\circ : \neg\neg B^\circ}{\Gamma^\circ, k : \neg(A^\circ \rightarrow \neg\neg B^\circ) \vdash \lambda x. t^\circ : A^\circ \rightarrow \neg\neg B^\circ}}{\Gamma^\circ, k : \neg(A^\circ \rightarrow \neg\neg B^\circ) \vdash k(\lambda x. t^\circ) : \perp} \quad \Gamma^\circ \vdash (\lambda x. t)^\circ = \lambda k. k(\lambda x. t^\circ) : \neg\neg(A^\circ \rightarrow \neg\neg B^\circ)$$

- t-app:

$$\frac{\frac{\Gamma^\circ, k : \neg B^\circ \vdash t^\circ : \neg\neg(A^\circ \rightarrow \neg\neg B^\circ)}{\Gamma^\circ, k : \neg B^\circ \vdash \lambda x. s^\circ(\lambda y. x y k) : \neg(A^\circ \rightarrow \neg\neg B^\circ)} \text{i.h. + weakening} \quad \frac{\frac{\frac{\Gamma^\circ, k : \neg B^\circ, x : A^\circ \rightarrow \neg\neg B^\circ \vdash s^\circ(\lambda y. x y k) : \perp}{\Gamma^\circ, k : \neg B^\circ \vdash \lambda x. s^\circ(\lambda y. x y k) : \neg(A^\circ \rightarrow \neg\neg B^\circ)} \text{by } \pi \text{ below}}{\Gamma^\circ, k : \neg B^\circ \vdash t^\circ(\lambda x. s^\circ(\lambda y. x y k)) : \perp} \text{i.h. + weakening}}{\Gamma^\circ \vdash (t s)^\circ = \lambda k. t^\circ(\lambda x. s^\circ(\lambda y. x y k)) : \neg\neg B^\circ}$$

where  $\pi$  is the following derivation:

$$\frac{\frac{\frac{\Delta, y : A^\circ \vdash x : A^\circ \rightarrow \neg\neg B^\circ}{\Delta, y : A^\circ \vdash x y : \neg\neg B^\circ} \quad \frac{\Delta, y : A^\circ \vdash y : A^\circ}{\Delta, y : A^\circ \vdash k : \neg B^\circ}}{\Delta, y : A^\circ \vdash x y k : \perp}}{\Delta \vdash \lambda y. x y k : \neg A^\circ}$$

□